PAMANTASAN NG LUNGSOD NG VALENZUELA
Tongco St., Maysan, Valenzuela City

COLLEGE OF ENGINEERING AND INFORMATION TECHNOLOGY
Department of Information Technology

SYLLABUS IN FUNDAMENTALS OF PROGRAMMING (LECTURE LABORATORY) CC 2

UNIVERSITY VISION STATEMENT	A dynamic center for the development of competent and competitive human resource as foundation for growth and advancement of the City of Valenzuela.	**UNIVERSITY MISSION STATEMENT**	To provide the citizens of Valenzuela an efficient and effective institution of higher learning that will make them skillful, productive, competent, civic-minded and God-loving toward a peaceful, healthy and progressive city.
COLLEGE VISION STATEMENT	Aims to become the premiere institution of higher learning providing the city with competent and committed engineers and IT professionals that will help the development of the city of Valenzuela and the nation.	**COLLEGE MISSION STATEMENT**	To develop the student of the city of Valenzuela into top caliber engineers and IT professionals who are proficient, committed and environmentally aware with good leadership skills that will comply with the needs of the city and the nation.
COLLEGE OBJECTIVES	**College Goals** Provide opportunities for qualified students to become professional engineers and technicians of the country. Specifically, the engineering programs are designed to: 1. Acquire full understanding of the scientific principles and knowledge of the particular field. 2. Develop a high level of competence in its engineering methods and applications. 3. Communicate effectively and succinctly important result of any technical study both verbally and technically. 4. Nurture the desire for continuing professional growth and explore new horizons in technology. 5. Imbue the graduates with socially and morally sound motivations and principles to live by. **College Core Values** The College is committed to: ● Academic excellence ● Integrity and professional leadership ● Scholarly research ● Commitment to service ● Life-long learning		

PROGRAM OUTCOMES (CMO. No. 25, s. 2015 Sec. 6.3.C)	

Graduate Attributes	Graduate Outcomes Code	Graduate Outcomes
Knowledge for Solving Computing Problems	IT01	Apply knowledge of computing, science, and mathematics appropriate to the discipline
	IT02	Understand best practices and standards and their applications
Problem Analysis	IT03	Analyze complex problems and identify and define the computing requirements appropriate to its solution.
	IT04	Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.
Design / Development of Solutions	IT05	Design, implement and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under carious constraints.
	IT06	Integrate IT-based solutions into the user environment effectively.
Modern Tool Usage	IT07	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession.
Individual and Team-Work	IT08	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish common goal.
	IT09	Assist in the creation of an effective IT project plan.
Communication	IT10	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.
Computing professionalism and Social Responsibility	IT11	Analyze the local and global impact of computing information technology on individuals, organizations, and society.
	IT12	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.
Life-Long Learning	IT13	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development.

COURSE INFORMATION			
COURSE TITLE	**Fundamentals of Computer Programming (Lecture and Laboratory)**	**COURSE CODE**	**CC 2**
COURSE PRE-REQUISITE	**None**	**CREDIT UNIT / CONTACT HOURS**	**3 Units** **5 hours / per week (18 Weeks, 90 hours)**

COURSE DESCRIPTION	The course "Fundamentals of Computer Programming" (CC102) provides an essential foundation for understanding the principles and concepts that underpin modern software development. This formal course is designed to introduce students to the fundamental aspects of programming and computational thinking, enabling them to develop logical problem-solving skills and create functional software solutions.
COURSE LEARNING OUTCOMES	Upon successful completion of the course, the students will be able to: 1. LO1. Design, implement, test, and debug a program, based on a given specification that uses each of the following fundamental programming components: (1) primitive data types, (2) basic computation, (3) simple I/O, (4) conditional and iterative structures, (5) definition of functions and parameters passing, and (6) recursion. 2. LO2. Analyze and simulate results of algorithms that may be implemented as a solution to a given problem Target: The desired standard is 100% with a passing grade of 75% or equivalent to 3.00 Target: The desired standard is 100% with a passing grade of 75% or equivalent to 3.00

ALIGNMENT OF COURSE OUTCOMES WITH SUMMATIVE ASSESSMENT TASK		
Course Outcomes	**Summative Assessment Tasks**	**Details**
1. Design, implement, test, and debug a program, based on a given specification that uses each of the following fundamental programming components: (1) primitive data types, (2) basic computation, (3) simple I/O, (4) conditional and iterative structures, (5) definition of functions and parameters passing, and (6) recursion. 2. Analyze and simulate results of algorithms that may be implemented as a solution to a given problem.	1. Written Output ● Quizzes ● Program Documentation 2. Non- Written Output ● Machine Problems ● Laboratory Exercises ● Recitation 3. Midterm / Final Examination	The summative tasks are graded requirements. They translate the outcomes of the course into what students are able to do given the breadth and depth of content knowledge.

	PROGRAM OUTCOMES AND RELATIONSHIPS TO THE PROGRAM EDUCATIONAL OBJECTIVES			
	PROGRAM OUTCOMES	**PROGRAM EDUCATIONAL OBECTIVES**		
		PG 1	**PG 2**	**PG 3**
IT01	Apply knowledge of computing, science, and mathematics appropriate to the discipline			✓
IT02	Understand best practices and standards and their applications		✓	
IT03	Analyze complex problems and identify and define the computing requirements appropriate to its solution.		✓	
IT04	Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.		✓	

IT05	Design, implement and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under carious constraints.			✓
IT06	Integrate IT-based solutions into the user environment effectively.		✓	
IT07	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession.			✓
IT08	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish common goal.		✓	
IT09	Assist in the creation of an effective IT project plan.	✓		
IT10	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.		✓	
IT11	Analyze the local and global impact of computing information technology on individuals, organizations, and society.			✓
IT12	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.		✓	
IT13	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development.		✓	

COURSE MAP

LEARNING OUTCOMES	PROGRAM OUTCOMES												
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**
LO1. Design, implement, test, and debug a program, based on a given specification that uses each of the following fundamental programming components: (1) primitive data types, (2) basic computation, (3) simple I/O, (4) conditional and iterative structures, (5) definition of functions and parameters passing, and (6) recursion.	**I**						**I**		**I**				
LO2. Analyze and simulate results of algorithms that may be implemented as a solution to a given problem	**I**						**I**		**I**				

Legend:

I: Achieve basic competency and develop a conceptual framework for more complex topics; an **introductory** course to an outcome.
E: Ability to apply knowledge and skills in complex situations; a course that **enabling** the outcome.
D: Showcase ability to interpret and apply what the students have learned; a course **demonstrating** an outcome

COURSE LEARNING PLAN						
TIME ALLOTMENT	**COURSE CONTENTS**	**ESSENTIAL QUESTION/S**	**INTENDED LEARNING OUTCOMES (ILOs)**	**SUGGESTED TEACHING /LEARNING ACTIVITIES**	**SUGGESTED ASSESSMENT TASKS**	**REFERENCES / TEACHING-LEARNING MATERIALS**
Week 1/ 5 hours	**Class Orientation** **Introduction to Computer Programming** Programming language. History of Java. Program development Process / Phases of programming	1. *What Is Computer Programming?* 2. *How computer programs work?* 3. *What is a programming language?* 4. *How Do We Write Code?*	Define basic programming terminology	○ Lecture / Discussion ○ Installing the Integrated Development Environment (IDE)	Installing and Launching the IDE that will be use in the following discussion. Short quiz regarding the discussion.	Savitch, W. (2019). Java An Introduction to Problem Solving and Programming (Mock, K., Horton, M., Partridge, J., Maheshwari, S., Mitra, A., Kumar, V., & Pellegrino, C.)(Eight Edition). Pearson Education Limited Introduction to JAVA Programming 8th Edition
Weeks 2/ 5 hours	**Anatomy of a Basic Program Source Code** Creating a java project Naming conventions Java - Basic Syntax Data types Variable Declaration	1. *What is a source code?* 2. *What are the primitive data types of Java?* 3. *How to declare and assign variables?*	Review and validate the concepts, ideas, events, and figures previously discussed in Introduction to Computer Programming Create and run a computer program using the knowledge acquired in the discussion.	○ Lecture / Discussion ○ Coding Demonstration ○ Printing using Java ○ Practice coding using different Java data types ○ Exercise/ Task or Machine problems	Fill-in-the blanks, Coding exercise (Quiz) Machine problems involving basic operations. Laboratory exercises involving basic operations. (Assignment)	Savitch, W. (2019). Java An Introduction to Problem Solving and Programming (Mock, K., Horton, M., Partridge, J., Maheshwari, S., Mitra, A., Kumar, V., & Pellegrino, C.)(Eight Edition). Pearson Education Limited Introduction to JAVA Programming 8th Edition
Week 4-5 10 hours	**Java Basic Operators** Escape Characters Assignment Operator Arithmetic Operators Relational Operator Boolean Logical Operators	1. *What are the basic operators of Java?* 2. *What is the use of these operators?*	Create and run a computer program using the knowledge acquired in the discussion.	○ Lecture / Discussion ○ Coding Demonstration ○ Performing standard arithmetic operators ○ Practice coding using the basic Operations	Laboratory exercises involving basic operations. (Assignment) Machine problems involving basic operations.	Savitch, W. (2019). Java An Introduction to Problem Solving and Programming (Mock, K., Horton, M., Partridge, J., Maheshwari, S., Mitra, A., Kumar, V., & Pellegrino, C.)(Eight Edition). Pearson Education Limited

				○ Exercise/ Task or Machine problems		
Week 5/ 5 hours	**Conditional Structures 1** Flowchart If statement if … else statement Using Multiple Statements in an if or if … else Structure Using Switch Statements	*1. What is the essence of conditional structures?* *2. What are the types of conditional structures?* *3. How to use Multiple Statements in an if or if … else Structure*	Define what is Conditional structure in Java Plan decision-making logic and Make decisions with the if and if … else structures	○ Lecture / Discussion ○ Coding Demonstration ○ Exercise/ Task or Machine problems ○ Practice coding using the conditional structures	Machine problems involving conditional structures Laboratory exercises involving basic operations. (Assignment)	Sanderson, S. (2016). Java for Beginners: A Simple Start To Java Programming (Written By A Software Introduction to JAVA Programming 8th Edition
Week 6/ 5 hours	**Conditional Structure 2** Nesting if and if … else Statements Using the Conditional and NOT Operators Understanding Operator Precedence	*1. When to use a certain type of Conditional structure?* *2. How to use Nesting If and if else in Java?* *3. How to use Logical AND and OR Operators*	Define what is Nested if and if else in Java Making Accurate and Efficient Decisions Learn how to use the Logical AND and OR Operators	○ Lecture / Discussion ○ Coding Demonstration ○ Exercise/ Task or Machine problems ○ Making Accurate and Efficient Decisions ○ Practice coding using the conditional statement if, if else and switch.	Machine problems involving conditional Structure. Laboratory exercises involving Conditional Structure. (Assignment) Quiz about the conditional structures	Sanderson, S. (2016). Java for Beginners: A Simple Start To Java Programming (Written By A Software Introduction to JAVA Programming 8th Edition

Week 7/ 5 hours	**Iteration Structures 1** Creating a While Loop Creating a for Loop Creating a do while loop Introduction to Nested For Loops Learning How and When to a looping statement	1. *What is an Iteration structure?* 2. *What are the types of Iteration structures* 3. *What is a Nested For loop?* 4. *How do we write a nested For loop?*	Define what is an Iteration structure Learning About the Loop Structure and creating loops Discussion about loop structure and Nested Loop	○ Lecture / Discussion ○ Coding Demonstration ○ Exercise/ Task or Machine problems ○ Making Accurate and Efficient Looping ○ Practice using all the looping structures in Java	Guessing the output every Iteration. (exercise) Laboratory exercises involving Iteration Structure. Creating a program that uses Iteration Structure (Assignment)	Sanderson, S. (2016). Java for Beginners: A Simple Start To Java Programming (Written By A Software Introduction to JAVA Programming 8th Edition
Week 8/ 5 hours	**Iteration Structures 2** Introduction to Indefinite loop Difference between a definite and indefinite loop Learning how to improve the loop performance	1. *What is an Indefinite loop?* 2. *How do we write an Indefinite loop?* 3. *How can we improve the loop performance?*	Differentiating a Definite Loop and Indefinite Loop. Learning how to improve the Loop Structure	○ Lecture / Discussion ○ Coding Demonstration ○ Exercise/ Task or Machine problems ○ Making Accurate and Efficient Looping ○ Practice using all the looping structures in Java	Laboratory exercises involving Iteration Structure. Creating programs that prints pattern outputs using looping (Laboratory) Machine problems involving Iteration Structures.	Sanderson, S. (2016). Java for Beginners: A Simple Start To Java Programming (Written By A Software Introduction to JAVA Programming 8th Edition
Week 9	**Midterm Examination**					
Week 10/ 5 hours	**Characters, Strings, and the StringBuilder** Understanding String Data Problems Manipulating Characters Using other String Method	1. *How to Compare Strings?* 2. *How to manipulate a character?*	Understanding how to manipulate characters and Strings Learning how to Declare and Compare String Objects Learning the functions of String Methods	○ Lecture / Discussion ○ Coding Demonstration ○ Exercise/ Task or Machine problems ○ Creating programs that involves String and Characters	Laboratory exercises involving String. Creating a program that uses Strings and Characters (Assignment) Quiz about the conditional structures-Characters and Strings	Sanderson, S. (2016). Java for Beginners: A Simple Start To Java Programming (Written By A Software Introduction to JAVA Programming 8th Edition

Week 11-12/ 10 hours	**Array** Declare and initialize an array Search an array Pass arrays to and return arrays from methods	*1. What is an Array?* *2. How do we declare and Initialize an Array?* *3. How to Pass an Arrays to a Methods?*	Learning the basics of array Understanding how to pass Array in a method Learning how to sort and search in arrays	○ Lecture / Discussion ○ Coding Demonstration ○ Exercise/ Task or Machine problems ○ Creating programs that involves Array	Guessing the output base on the given code (Quiz) Machine problems involving arrays. (Face to Face Laboratory) Laboratory exercises involving arrays.	Savitch, W. (2019). Java An Introduction to Problem Solving and Programming (Mock, K., Horton, M., Partridge, J., Maheshwari, S., Mitra, A., Kumar, V., & Pellegrino, C.)(Eight Edition). Pearson Education Limited Introduction to JAVA Programming 8th Edition
Week 13-14/ 10 hours	**Methods** Static Variables and Methods Closer look at argument passing Introduction to access control Types of methods	*1. What is a method?* *2. What are the different types of methods?* *3. What is parameter passing in method?*	Understanding what is a method and the different types of methods Understanding how methods passed and returned a value Learning about access controls	○ Lecture / Discussion ○ Coding Demonstration ○ Exercise/ Task or Machine problems ○ Creating programs that involves passing arguments to methods	Machine problems involving methods. Laboratory exercises involving Methods and passing data in to methods. Quiz about methods and types of methods	Savitch, W. (2019). Java An Introduction to Problem Solving and Programming (Mock, K., Horton, M., Partridge, J., Maheshwari, S., Mitra, A., Kumar, V., & Pellegrino, C.)(Eight Edition). Pearson Education Limited Introduction to JAVA Programming 8th Edition
Week 15/ 5 hours	**Array List** Using the ArrayList class Adding items to ArrayList	*1. What is an array list?* *2. How array list differs from an array?*	Understanding the Array List Class of Java Learning how to use methods of ArrayList Understanding the Limitations of the ArrayList Class	○ Lecture / Discussion ○ Coding Demonstration ○ Exercise/ Task or Machine problems ○ Creating programs that involves Array List	Machine problems involving array lists. Laboratory exercises involving array lists. Laboratory exercises involving array lists.	Sanderson, S. (2016). Java for Beginners: A Simple Start To Java Programming (Written By A Software Engineer). Globalized Healing, LLC Introduction to JAVA Programming 8th Edition

Week 16-17/ 10 hours	**Exception Handling** Try code and Catch exception Try with resources Throw and catch multiple Exceptions Manual throw User-define exceptions Use the finally block	1. *What is exception handling?* 2. *What are the advantages of using excetion handling?*	Learning about exception Handling Understanding the advantages of exception Handling	○ Lecture / Discussion ○ Coding Demonstration ○ Exercise/ Task or Machine problems ○ Creating programs that involves Exception Handling	Laboratory exercises involving Exception Handling. Quiz about Exception Handling	Sanderson, S. (2016). Java for Beginners: A Simple Start To Java Programming (Written By A Software Engineer). Globalized Healing, LLC Introduction to JAVA Programming 8th Edition
Week 18	**Final Term Examination**					

COURSE REFERENCES	
REFERENCES	1. Daniel Liang (2011) Introduction to JAVA Programming 8th Edition: Pearson Publishing 2. Farell, J. (2012). Java Programming, Sixth Edition (Marie Lee)(Sixth Edition). Course Technology 3. Sanderson, S. (2016). Java for Beginners: A Simple Start To Java Programming (Written By A Software Engineer). Globalized Healing, LLC 4. Savitch, W. (2019). Java An Introduction to Problem Solving and Programming (Mock, K., Horton, M., Partridge, J., Maheshwari, S., Mitra, A., Kumar, V., & Pellegrino, C.)(Eight Edition). Pearson Education Limited 5. Schildt, H. (2019). Java: The Complete Reference (Coward, D.)(Eleventh Edition). Globalized Healing, LLC
OTHER SUPPORT MATERIALS	1. Simplilearn. (2022). Complete Java Full Course for 2022 | Java Tutorial for Beginners | Java Programming | Simplilearn. [Video]. https://youtu.be/7yf6676xMhM 2. freeCodeCamp.org. (2018) Marcus Biel. Learn Java 8 - Full Tutorial for Beginners [Video]. https://youtu.be/grEKMHGYyns

COURSE REQUIREMENTS AND POLICIES			
GRADING SYSTEM	Midterm / Final Examination **40%** Written Outputs **30%** Performance Based Outputs **30%** **100%**	Midterm grade + Final term grade 2 = **Final grade**	**Number System Grading** 1.00 = 97-100 1.25 = 94-96 1.50 = 91-93 1.75 = 88-90 2.00 = 85-87 2.25 = 82-84 2.50 = 79-81 2.75 = 76-78 3.00 = 75 5.00 = Failed
POLICIES (as agreed by the class)	1. Attendance will be checked at the start of each session. Students arriving 15 minutes after the start of the official class hours will be given a mark of late. Three marks for late is automatically converted to a mark of absent. When students incur absences exceeding 20% of the total class hours for the term in a course, they will be automatically given a failing grade of 5.00 (PLV Student Manual p. 24). 2. Lectures will cover fundamental concepts presented in the reference books, instructional materials and websites provided. It is strongly recommended and assumed that students had read the relevant text sections while the course progresses. 3. Students can take photos of class lectures, materials or equipment; however, they are prohibited to record videos and snippets. 4. Activities, plates, seatwork and homework must be completed and submitted on the due date specified. Late submissions, if reasons are valid, may be accepted upon the discretion of the instructor. 5. Quizzes, exam results, seatwork and homework will be returned to the students a week after submission of such activities.		

	6. All quizzes and major examinations are announced. Special quizzes will not be entertained regardless of reasons. Major examinations missed due to valid reasons and with proper documentation are graded zero by default unless a special major exam is scheduled. Only one major examination can be waived for special schedule. Missed or failed major exams and quizzes are not subjected for substitution for another activity. 7. There is no removal examination for this course. 8. University rules on cheating and plagiarism will be applied.

INSTRUCTORS INFORMATION		
Name & Signature	Consultation Schedule	Email Address
Rosalie P. Alejandro, LPT, MIT	5:30PM-8:30PM | Saturday	ceitalejandrorosalie@gmail.com

Date Last Revised: **January 31, 2025**	**Endorsed by:** **Kenmar C. Bernardino** Chairperson, Information Technology Department
Revision Date: **August 12, 2024** **August 22, 2023**	**Recommending Approval:** **Engr. Jordan N. Velasco** Dean, College of Engineering and Information Technology
	Approved by: **Dr. Nelda Gene C. Mariano** Vice President for Academic Affairs